

Ph134, Lecture 15

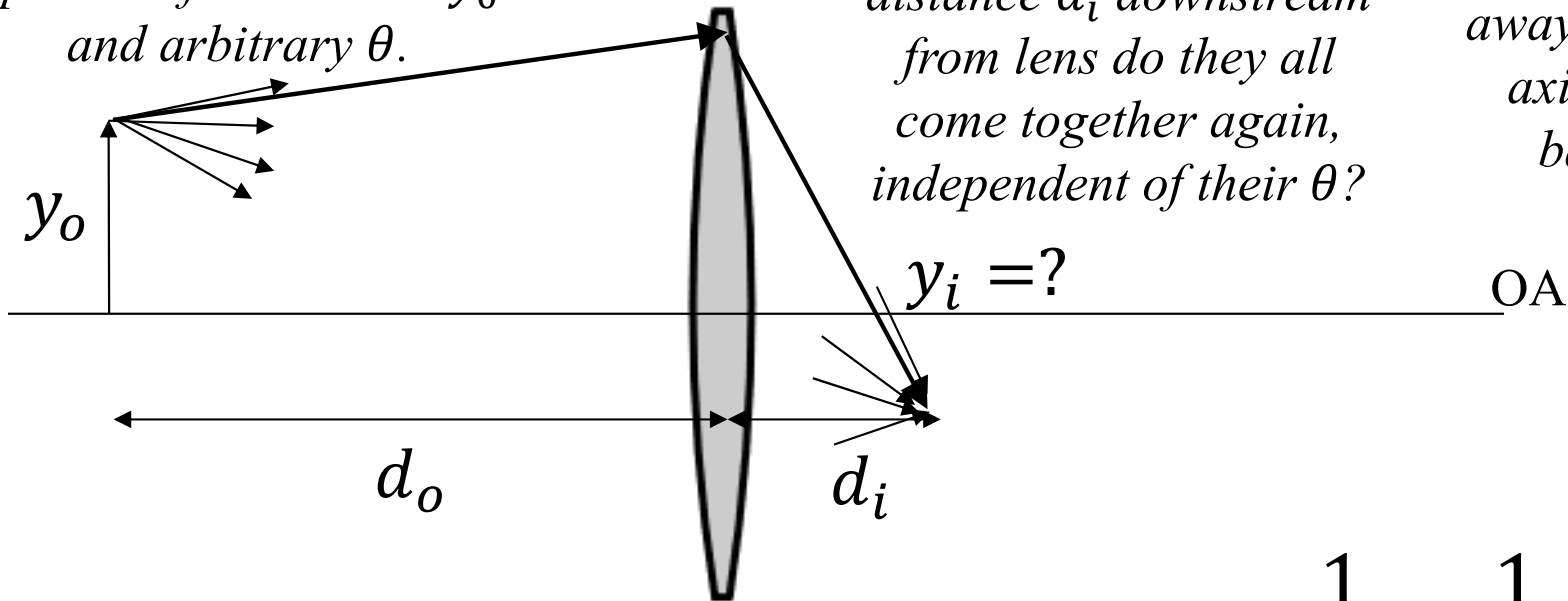
- **Thin lens equation and meaning of focal length**
- **Imaging with systems of lenses:**
 - **Real and virtual images**
 - **Real and virtual objects**
 - **It's all about how the rays are pointing!**
- **Issues with lensing:**
 - **Chromatic aberration**
 - **Monochromatic aberrations from imprecision of paraxial approximation**

Thin lens equation and magnification:

Consider rays that start
longitudinal distance d_o
upstream from lens at y_o
and arbitrary θ .

At what longitudinal
distance d_i downstream
from lens do they all
come together again,
independent of their θ ?

At what position
away from the optical
axis do they come
back together?



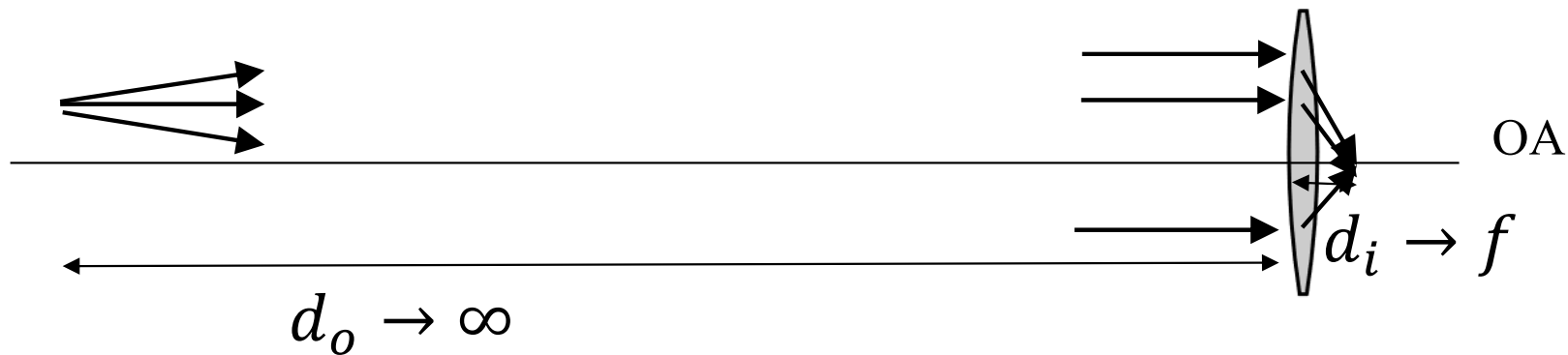
$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Magnification $\frac{y_i}{y_o} = -\frac{d_i}{d_o}$

Is the focal point the place where the lens forms a focused image? Only if the object being imaged is infinitely far away!

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

An object at infinite distance will be imaged at f downstream from the lens.



Or: an incoming collimated beam (a beam with near-planar wavefronts) will be focused to a spot at f downstream from the lens.

Virtual images:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

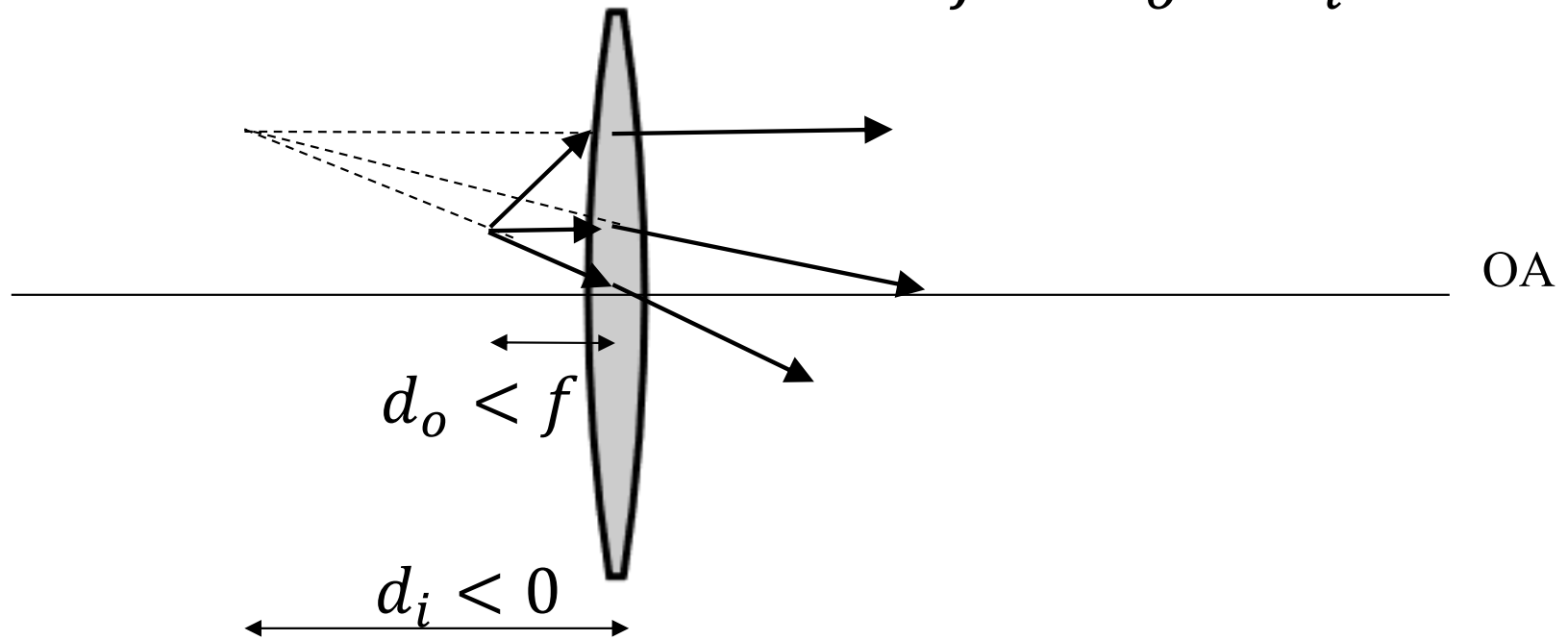
*An object at infinite distance will be imaged
at f downstream from the lens.*



*Nothing at the image position, no
light propagating “back” to it.
But rays downstream of lens look
like they came from that spot.*

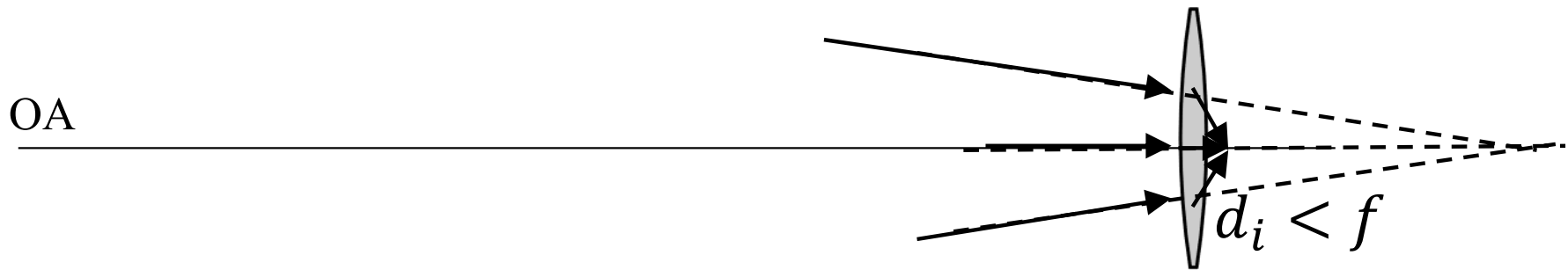
Do converging lenses make virtual images?

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$



*Nothing at the image position, no light propagating “back” to it.
But rays downstream of lens look like they came from that spot.*

Can we put the object farther away than ∞ ? (Converging lens case)



$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$d_o < 0$$

*Rays incident on lens are already converging – on path as if coming from object at a position DOWNSTREAM of the lens.
Need another converging lens between actual object and this lens to get this situation.*

Real and virtual images:

Real image: downstream of lens. $d_i > 0$

Light rays coming from lens actually pass through image pos'n.

Virtual image: upstream of lens. $d_i < 0$

Light rays coming from lens are angled as if they pass through image pos'n.

Real and virtual objects:

Real object: upstream of lens. $d_o > 0$

Light rays incident on lens actually come directly from each point on the object.

Virtual object: downstream of lens. $d_o < 0$

Light rays incident on lens are angled as if they are converging on (or if time reversed, diverging from) the "object."

The human eye and corrective lenses:

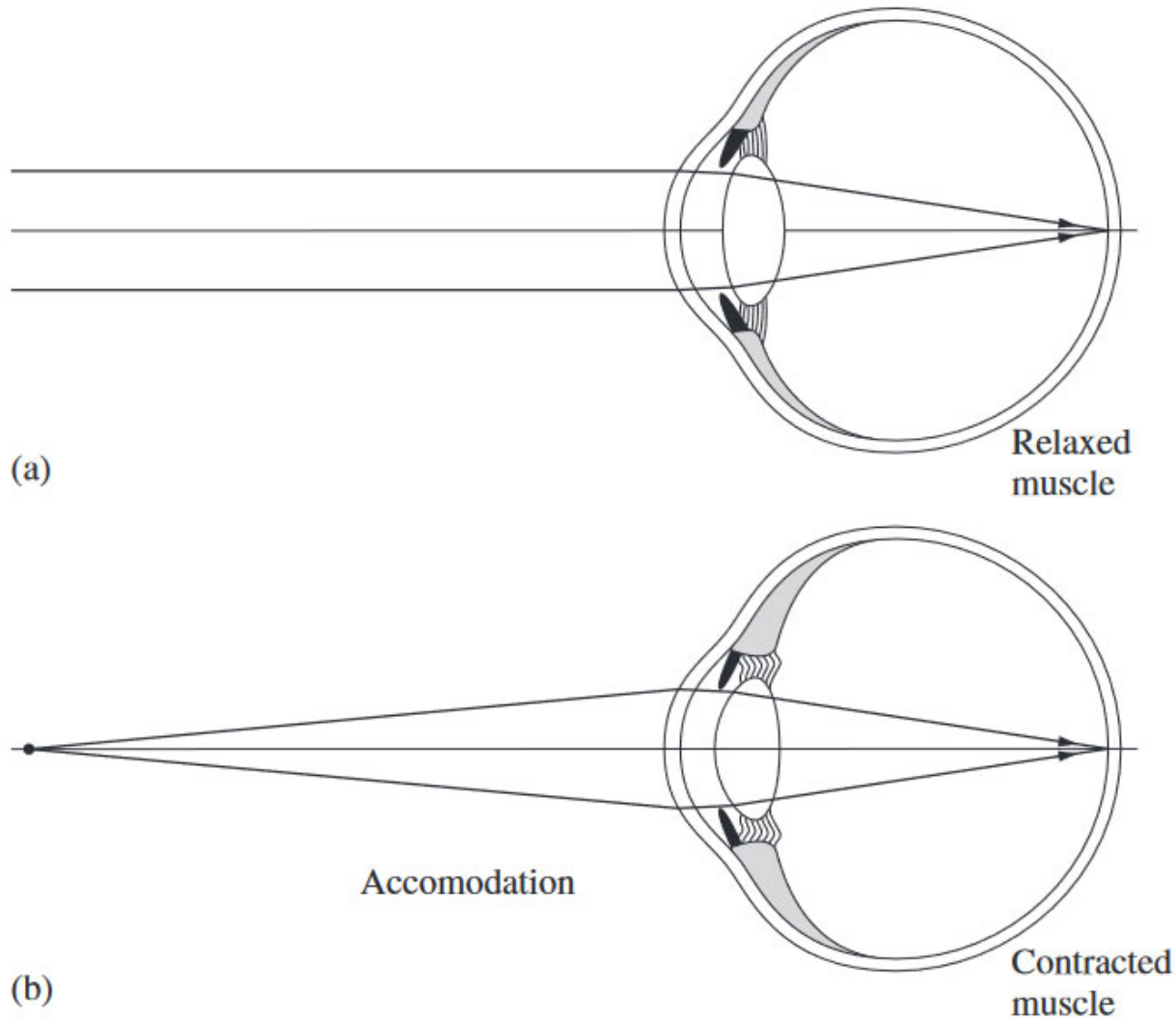
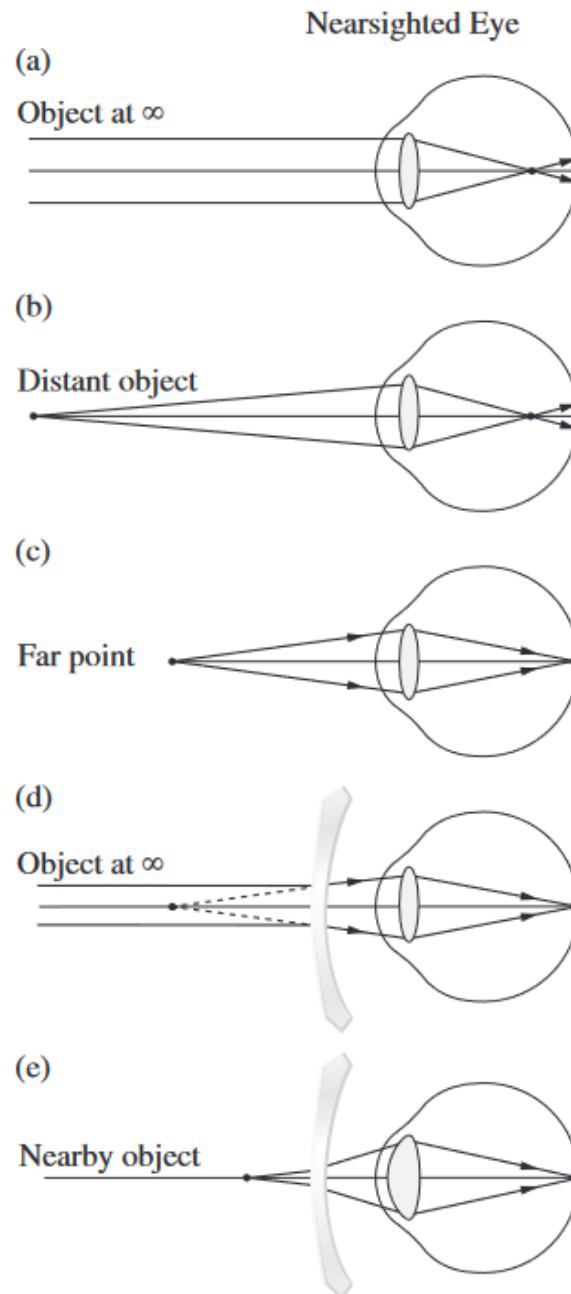


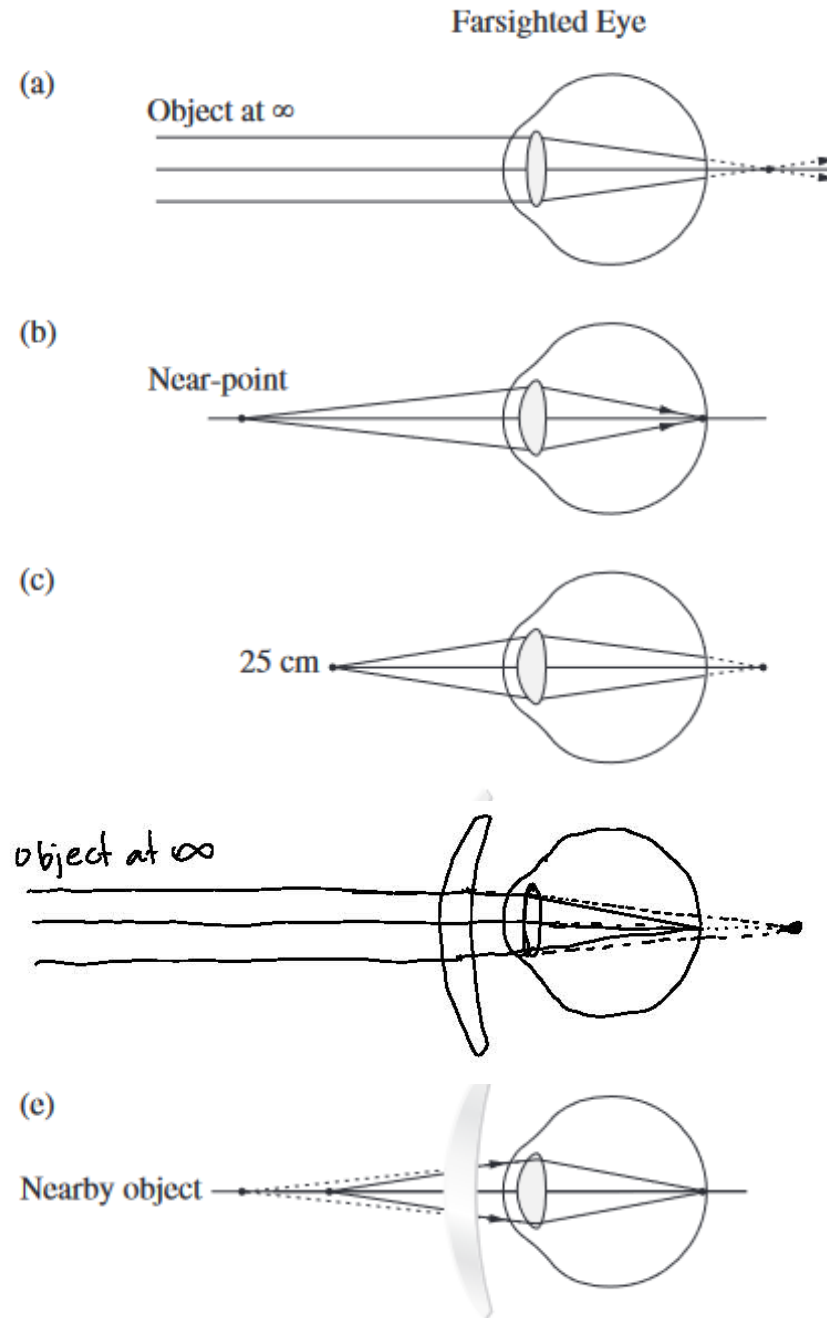
Figure 5.93 Accommodation—changes in the lens configuration.

Nearsighted (can only see objects at “far point” and closer):



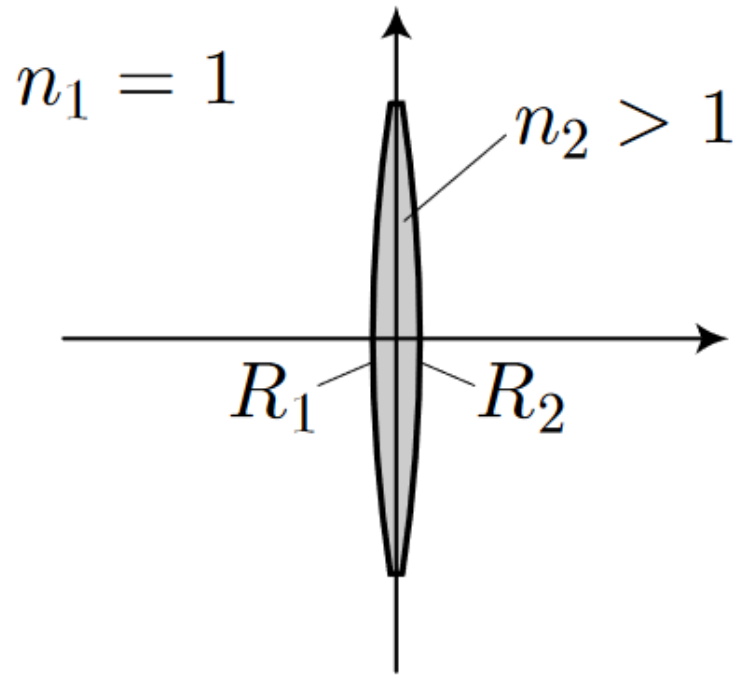
Figures from Hecht, Optics

Farsighted (can only see objects at “near point” and farther):



Figures adapted from Hecht, Optics

Chromatic aberration:



$$\frac{1}{f} = (n_2 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

*If n depends on wavelength,
 f depends on wavelength.*



<https://photographylife.com/what-is-chromatic-aberration>

Achromatic lenses:

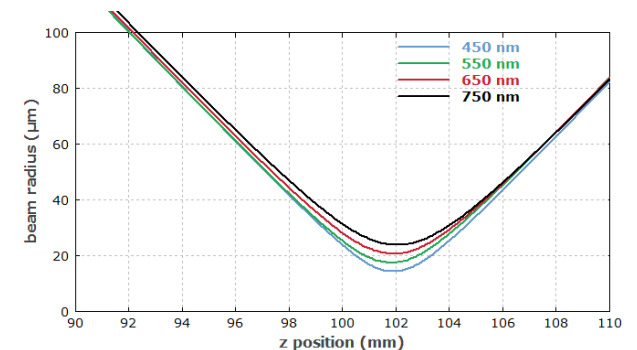
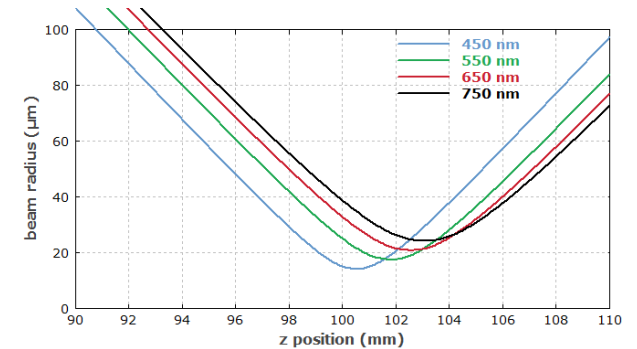
← → ⌂ ↻ — 100% + 🔒 www.rp-photonics.com/achromatic_optics.html

For improving the situation, we can now use an achromatic lens doublet, consisting of two parts (see Figure 2): a biconvex lens made of BK7 and concave-convex lens made of SF2 (a **flint glass**). The basic idea is to realize an overall focusing lens by combining a strongly focusing and a less strongly defocusing lens, where the latter exhibits stronger chromatic dispersion. Although the defocusing effect of the second lens is weaker than the focusing effect of the first one, its chromatic aberrations can compensate those of the other lens.

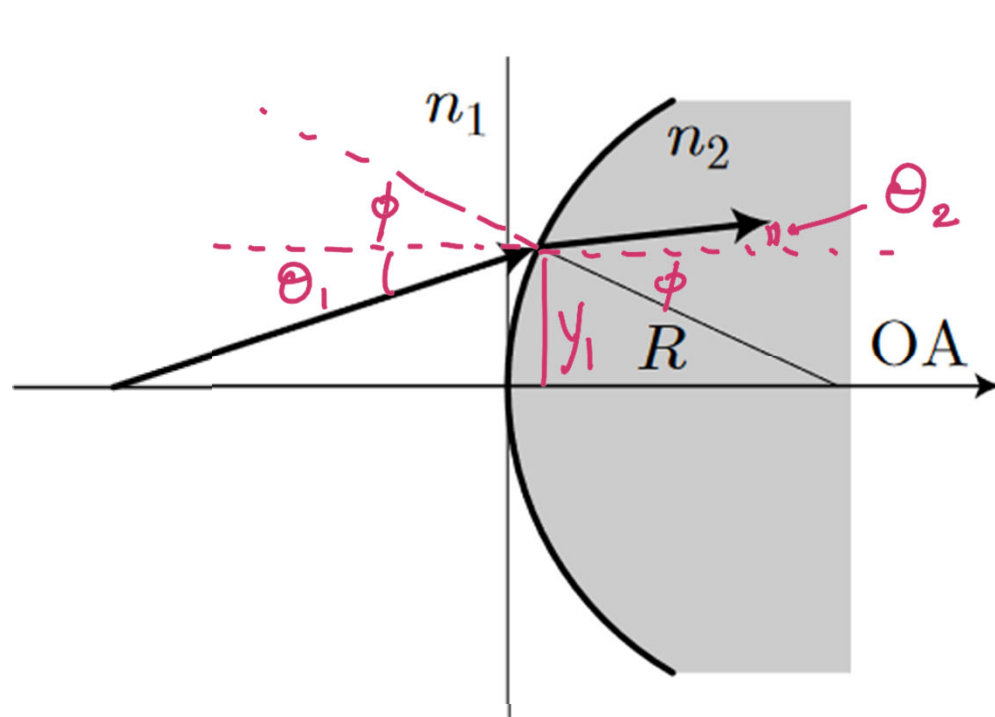
The considered doublet lens design has been numerically optimized with the **RP Resonator software** such that we obtain a **focal length** close to 100 mm (measured from the middle of the lens at $z = 2$ mm) in the wavelength range from 400 nm to 800 nm. The curvature radius on the left side



Figure 2: Design of an achromatic lens doublet. The left part is made from a **crown glass**, and the right one from a **flint glass**.



Monochromatic aberrations due to nonideality of spherical shape and/or imprecision of the paraxial approximation:



$$y_2 = y_1$$

$$n_1 \sin(\theta_1 + \phi) = n_2 \sin(\theta_2 + \phi)$$

$$n_1(\theta_1 + \phi) \approx n_2(\theta_2 + \phi)$$

$$n_1\left(\theta_1 + \frac{y_1}{R}\right) \approx n_2\left(\theta_2 + \frac{y_1}{R}\right)$$

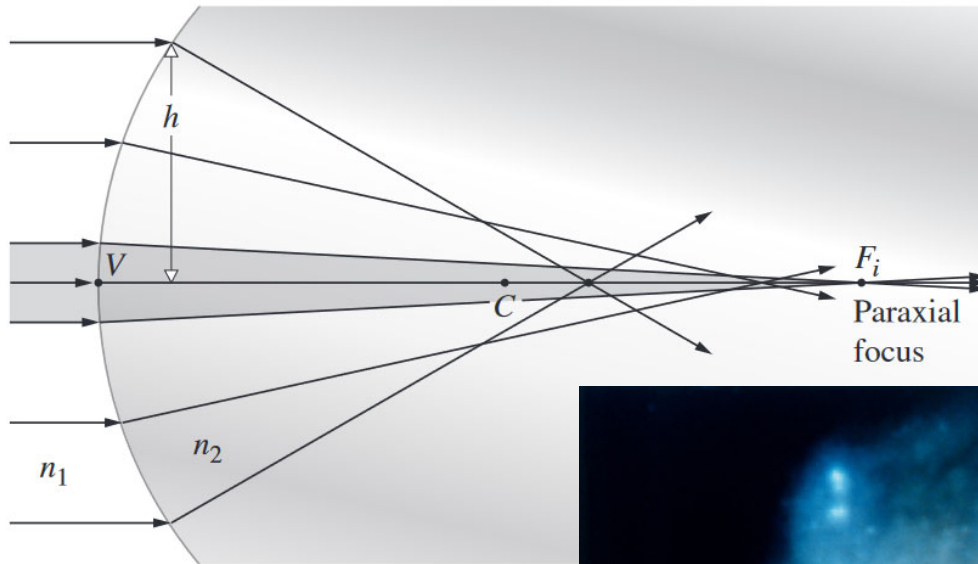
$$\theta_2 \approx \left(\frac{n_1 - n_2}{n_2}\right) \frac{1}{R} y_1 + \frac{n_1}{n_2} \theta_1$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -\frac{(n_2 - n_1)}{n_2 R} & \frac{n_1}{n_2} \end{bmatrix}$$

convex (shown): $R > 0$

concave: $R < 0$

Spherical aberration and more:



from Hecht, Optics



HST images of the M-100 galaxy with (before repair) and without (after repair) spherical aberration. (NASA)

<https://micro.magnet.fsu.edu/primer/lightandcolor/lenseshome.html>

https://www.rp-photonics.com/spherical_aberrations.html